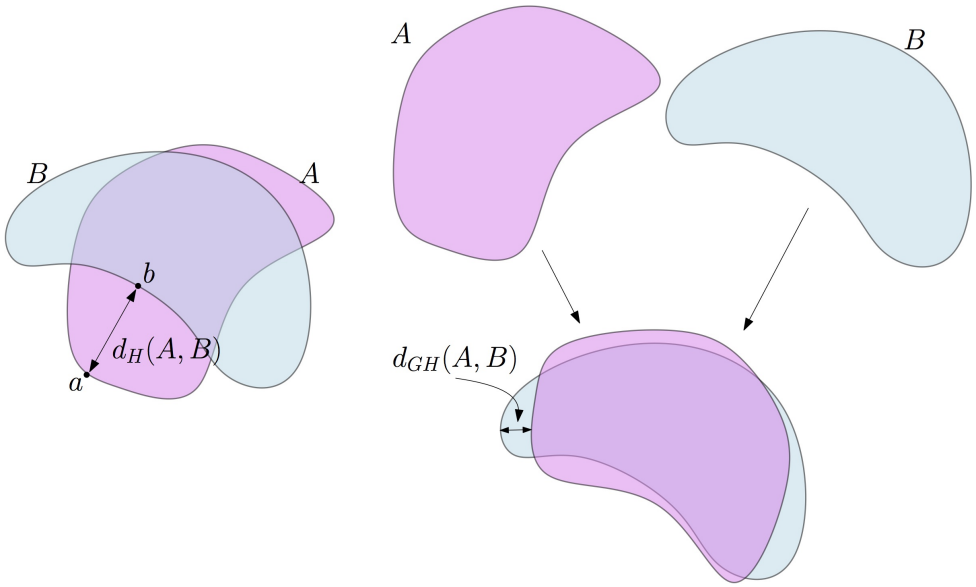

Figure 1: Right: the Hausdorff distance between two subsets A and B of the plane. In this example, $d_H(A,B)$ is the distance between the point a in A which is the farthest from B and its nearest neighbor b on B. Left: The Gromov-Hausdorff distance between A and B. A can been rotated - this is an isometric embedding of A in the plane - to reduce its Hausdorff distance to B. As a consequence, $d_{GH}(A,B) \leq d_H(A,B)$.

It is a basic and classical result that the Hausdorff distance is indeed a distance on the set of compact subsets of a metric space. From a TDA perspective it provides a convenient way to quantify the proximity between different data sets issued from the same ambient metric space. However, it sometimes occurs in that one has to compare data set that are not sampled from the same ambient space. Fortunately, the notion of Hausdorff distance can be generalized to the comparison of any pair of compact metric spaces, giving rise to the notion of *Gromov-Hausdorff distance*.

Two compact metric spaces (M_1, ρ_1) and (M_2, ρ_2) are *isometric* if there exists a bijection $\phi : M_1 \to M_2$ that preserves distances, i.e. $\rho_2(\phi(x), \phi(y)) = \rho_1(x, y)$ for any $x, y \in M_1$. The Gromov-Hausdorff distance measures how far two metric space are from being isometric.

Definition 1. *The Gromov-Haudorff distance $d_{GH}(M_1, M_2)$ between two compact metric spaces is the infimum of the real numbers $r \geq 0$ such that there exists a metric space (M, ρ) and two compact subspaces $C_1, C_2 \subset M$ that are isometric to M_1 and M_2 and such that $d_H(C_1, C_2) \leq r$.*

The Gromov-Hausdorff distance will be used later, in Section 5, for the study of stability properties persistence diagrams.

Connecting pairs of nearby data points by edges leads to the standard notion of neighboring graph from which the connectivity of the data can be analyzed, e.g. using some clustering algorithms. To go beyond connectivity, a central idea in TDA is to build higher dimensional equivalent of neighboring graphs by not only connecting pairs but also $(k+1)$-uple of nearby data points. The resulting objects, called simplicial complexes, allow to identify new topological features such as cycles, voids and their higher dimensional counterpart.

Geometric and abstract simplicial complexes. Simplicial complexes can be seen as higher dimensional generalization of graphs. They are mathematical objects that are both topological and combinatorial, a property making them particularly useful for TDA.

Given a set $\mathbb{X} = \{x_0, \cdots, x_k\} \subset \mathbb{R}^d$ of $k+1$ affinely independent points, the *k-dimensional simplex* $\sigma = [x_0, \cdots x_k]$ spanned by $\mathbb{X}$ is the convex hull of $\mathbb{X}$. The points of $\mathbb{X}$ are called the

vertices of σ and the simplices spanned by the subsets of $\mathbb{X}$ are called the *faces* of σ. A *geometric simplicial complex* K in $\mathbb{R}^d$ is a collection of simplices such that:
i) any face of a simplex of K is a simplex of K,
ii) the intersection of any two simplices of K is either empty or a common face of both.

The union of the simplices of K is a subset of $\mathbb{R}^d$ called the underlying space of K that inherits from the topology of $\mathbb{R}^d$. So, K can also be seen as a topological space through its underlying space. Notice that once its vertices are known, K is fully characterized by the combinatorial description of a collection of simplices satisfying some incidence rules.

Given a set V, an *abstract simplicial complex* with vertex set V is a set $\tilde{K}$ of finite subsets of V such that the elements of V belongs to $\tilde{K}$ and for any $\sigma \in \tilde{K}$ any subset of σ belongs to $\tilde{K}$. The elements of $\tilde{K}$ are called the faces or the simplices of $\tilde{K}$. The dimension of an abstract simplex is just its cardinality minus 1 and the dimension of $\tilde{K}$ is the largest dimension of its simplices. Notice that simplicial complexes of dimension 1 are graphs.

The combinatorial description of any geometric simplicial K obviously gives rise to an abstract simplicial complex $\tilde{K}$. The converse is also true: one can always associate to an abstract simplicial complex $\tilde{K}$, a topological space $|\tilde{K}|$ such that if K is a geometric complex whose combinatorial description is the same as $\tilde{K}$, then the underlying space of K is homeomorphic to $|\tilde{K}|$. Such a K is called a *geometric realization* of $\tilde{K}$. As a consequence, abstract simplicial complexes can be seen as topological spaces and geometric complexes can be seen as geometric realizations of their underlying combinatorial structure. So, one can consider simplicial complexes at the same time as combinatorial objects that are well-suited for effective computations and as topological spaces from which topological properties can be inferred.

Building simplicial complexes from data. Given a data set, or more generally a topological or metric space, there exist many ways to build simplicial complexes. We present here a few classical examples that are widely used in practice.

A first example, is an immediate extension of the notion of α-neighboring graph. Assume that we are given a set of points $\mathbb{X}$ in a metric space (M, ρ) and a real number $\alpha \geq 0$. The *Vietoris-Rips complex* $\mathrm{Rips}_\alpha(\mathbb{X})$ is the set of simplices $[x_0, \ldots, x_k]$ such that $d_{\mathbb{X}}(x_i, x_j) \leq \alpha$ for all (i, j). It follows immediately from the definition that this is an abstract simplicial complex. However, in general, even when $\mathbb{X}$ is a finite subset of $\mathbb{R}^d$, $\mathrm{Rips}_\alpha(\mathbb{X})$ does not admit a geometric realization in $\mathbb{R}^d$; in particular, it can be of dimension higher than d.

Closely related to the Vietoris-Rips complex is the *Čech complex* $\mathrm{Cech}_\alpha(\mathbb{X})$ that is defined as the set of simplices $[x_0, \ldots, x_k]$ such that the $k+1$ closed balls $B(x_i, \alpha)$ have a non-empty intersection. Notice that these two complexes are related by

$$\mathrm{Rips}_\alpha(\mathbb{X}) \subseteq \mathrm{Cech}_\alpha(\mathbb{X}) \subseteq \mathrm{Rips}_{2\alpha}(\mathbb{X})$$

and that, if $\mathbb{X} \subset \mathbb{R}^d$ then $\mathrm{Cech}_\alpha(\mathbb{X})$ and $\mathrm{Rips}_{2\alpha}(\mathbb{X})$ have the same 1-dimensional skeleton, i.e. the same set of vertices and edges.

The nerve theorem. The Čech complex is a particular case of a family of complexes associated to covers. Given a *cover* $\mathcal{U} = (U_i)_{i \in I}$ of $\mathbb{M}$, i.e. a family of sets U_i such that $\mathbb{M} = \cup_{i \in I} U_i$, the *nerve of* $\mathcal{U}$ is the abstract simplicial complex $C(\mathcal{U})$ whose vertices are the U_i's and such that

$$\sigma = [U_{i_0}, \cdots, U_{i_k}] \in C(\mathcal{U}) \text{ if and only if } \bigcap_{j=0}^{k} U_{i_j} \neq \emptyset.$$

Given a cover of a data set, where each set of the cover can be, for example, a local cluster or a grouping of data points sharing some common properties, its nerve provides a compact and global combinatorial description of the relationship between these sets through their intersection patterns - see Figure 3.

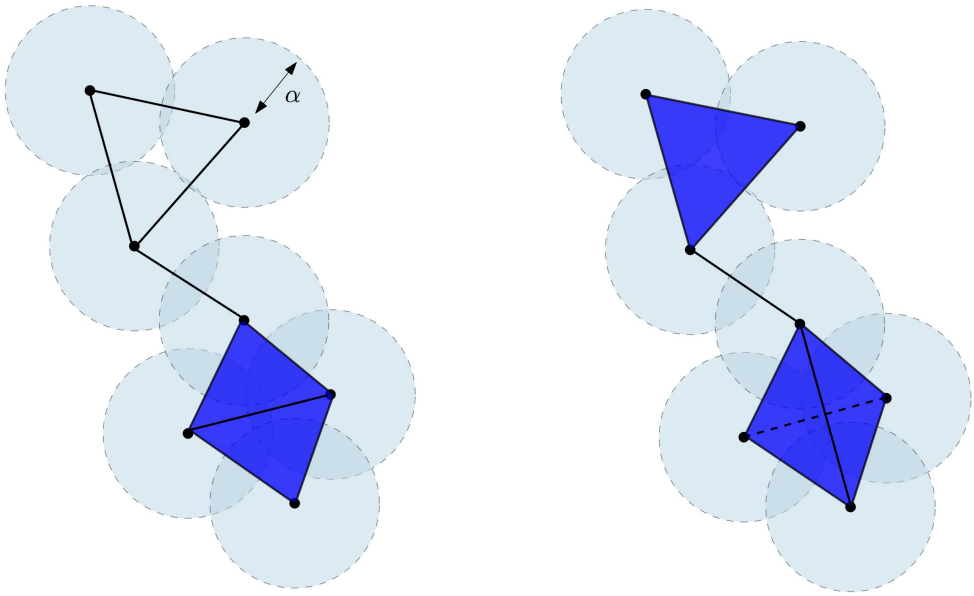

Figure 2: The Čech complex $\mathrm{Cech}_\alpha(\mathbb{X})$ (left) and the and Vietoris-Rips $\mathrm{Rips}_{2\alpha}(\mathbb{X})$ (right) of a finite point cloud in the plane $\mathbb{R}^2$. The bottom part of $\mathrm{Cech}_\alpha(\mathbb{X})$ is the union of two adjacent triangles, while the bottom part of $\mathrm{Rips}_{2\alpha}(\mathbb{X})$ is the tetrahedron spanned by the four vertices and all its faces. The dimension of the Čech complex is 2. The dimension of the Vietoris-Rips complex is 3. Notice that this later is thus not embedded in $\mathbb{R}^2$.

A fundamental theorem in algebraic topology, relates, under some assumptions, the topology of the nerve of a cover to the topology of the union of the sets of the cover. To be formally stated, this result, known as the Nerve Theorem, requires to introduce a few notions.

Two topological spaces X and Y are usually considered as being the same from a topological point of view if they are *homeomorphic*, i.e. if there exist two continuous bijective maps $f : X \to Y$ and $g : Y \to X$ such that $f \circ g$ and $g \circ f$ are the identity map of Y and X respectively. In many cases, asking X and Y to be homeomorphic turns out to be a too strong requirement to ensure that X and Y share the same topological features of interest for TDA. Two continuous maps $f_0, f_1 : X \to Y$ are said to be *homotopic* is there exists a continuous map $H : X \times [0,1] \to Y$ such that for any $x \in X$, $H(x,0) = f_0(x)$ and $H(x,1) = g(x)$. The spaces X and Y are then said to be *homotopy equivalent* if there exist two maps $f : X \to Y$ and $g : Y \to X$ such that $f \circ g$ and $g \circ f$ are homotopic to the identity map of Y and X respectively. The maps f and g are then called *homotopy equivalent*. The notion of homotopy equivalence is weaker than the notion of homeomorphism: if X and Y are homeomorphic then they are obviously homotopy equivalent, the converse being not true. However, spaces that are homotopy equivalent still share many topological invariant, in particular they have the same homology - see Section 4.
A space is said to be *contractible* if it is homotopy equivalent to a point. Basic examples of contractible spaces are the balls and, more generally, the convex sets in $\mathbb{R}^d$. Open covers whose all elements and their intersections are contractible have the remarkable following property.

Theorem 1 (Nerve theorem)**.** *Let $\mathcal{U} = (U_i)_{i \in I}$ be a cover of a topological space X by open sets such that the intersection of any subcollection of the U_i's is either empty or contractible. Then, X and the nerve $C(\mathcal{U})$ are homotopy equivalent.*

It is easy to verify that convex subsets of Euclidean spaces are contractible. As a consequence, if $\mathcal{U} = (U_i)_{i \in I}$ is a collection of convex subsets of $\mathbb{R}^d$ then $C(\mathcal{U})$ and $\cup_{i \in I} U_i$ are homotopy equivalent. In particular, if $\mathbb{X}$ is a set of points in $\mathbb{R}^d$, then the Čech complex $\mathrm{Cech}_\alpha(\mathbb{X})$ is homotopy equivalent to the union of balls $\cup_{x \in \mathbb{X}} B(x, \alpha)$.

The Nerve Theorem plays a fundamental role in TDA: it provide a way to encode the topology